

1. Which OSI layer is responsible for routing packets between networks?

- a) Data Link
- b) Network
- c) Transport
- d) Session

Answer: b

2. Which protocol provides reliable, connection-oriented delivery?

- a) UDP
- b) IP
- c) TCP
- d) ARP

Answer: c

3. What is the default port number for HTTP?

- a) 21
- b) 25
- c) 80
- d) 443

Answer: c

4. Which device connects two different networks and forwards packets based on IP?

- a) Hub
- b) Switch
- c) Router
- d) Repeater

Answer: c

5. IPv4 addresses are how many bits long?

- a) 16
- b) 32
- c) 64
- d) 128

Answer: b

6. Which protocol translates domain names to IP addresses?

- a) DHCP
- b) DNS
- c) FTP
- d) SNMP

Answer: b

7. Which topology connects every node to a single central cable?

- a) Star
- b) Bus
- c) Ring
- d) Mesh

Answer: b

8. Which protocol is used to send email?

- a) SMTP
- b) POP3
- c) IMAP
- d) HTTP

Answer: a

9. What does MAC stand for in networking?

- a) Media Access Control
- b) Memory Access Control
- c) Managed Address Control
- d) Media Address Code

Answer: a

10. Which layer of the TCP/IP model corresponds to the OSI Transport layer?

- a) Application
- b) Internet
- c) Transport
- d) Link

Answer: c

11. Explain the difference between TCP and UDP with one example use case for each.

(Write your answer in the space provided.)

12. Describe how a packet travels from a source host to a destination host on another network.

(Write your answer in the space provided.)